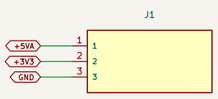
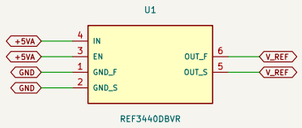
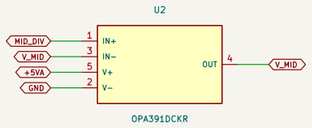
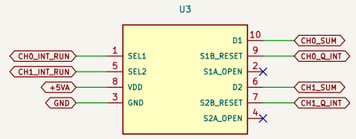
ARTIFICIAL ELECTRICAL LOADS ONLY / FIRST MONITORING SLICE

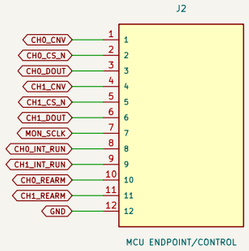
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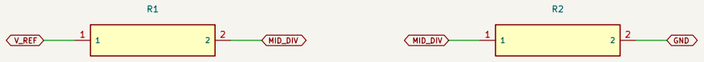


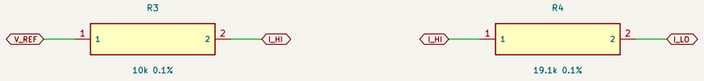


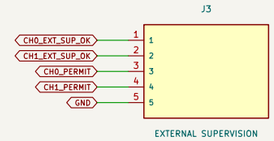


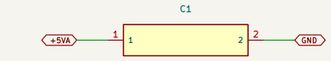


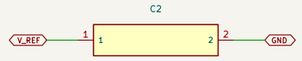


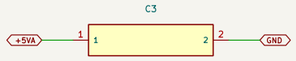


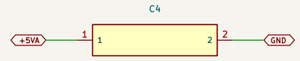


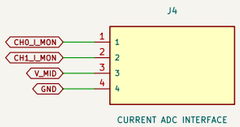
















V_REF=4.096V; V_MID=2.048V. Low INT_RUN resets; high opens reset for continuous signed integration.

I_LO=1.04757V / I_HI=3.04843V nominal (-+/-25.01mA). Threshold is NOT a proven complete-waveform cap.

EXT_SUP_OK must stay LOW until external rails/reference/watchdog/compliance/duration checks pass.

Two INA241 channels alone: up to32mW at5V. Input bias25..45uA/input is an unresolved leakage/bypass term.

Named global nets connect sheets; connectors remain explicit external design contracts. Read README before interpretation.

Source0_monitor

File: channel0.kicad_sch

Source1_monitor

File: channel1.kicad_sch

ARTIFICIAL LOADS ONLY; native schematic slice, not a completed implant

Project ATO

Sheet: /

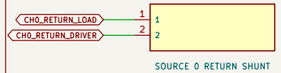
File: monitoring_cell.kicad_sch

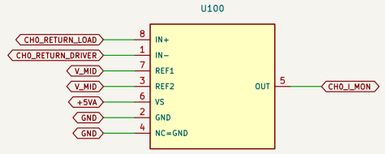
Title: ATO S4 / Shared reference and interfaces

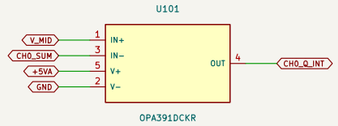
Size: A3	Date: 2026-10-12	**Rev: 0.2 DCK**
KiCad E.D.A. 10.0.6		Id: 1/3

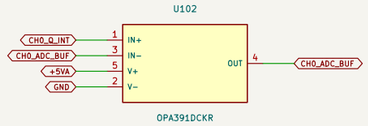
ARTIFICIAL ELECTRICAL LOADS ONLY / FIRST MONITORING SLICE

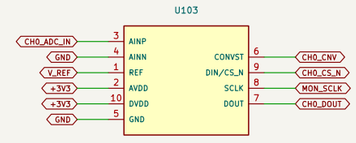
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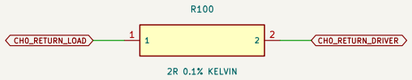








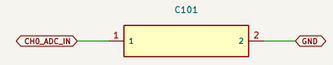


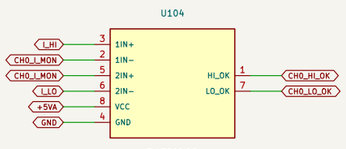


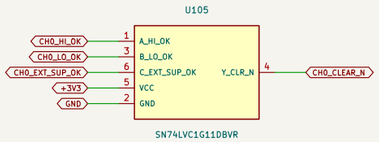


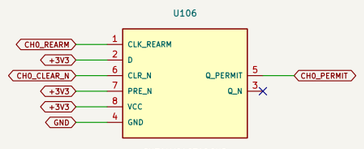


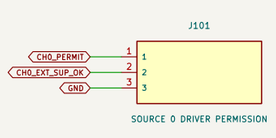


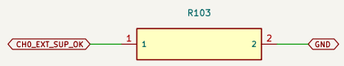


















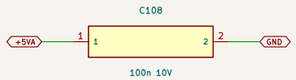


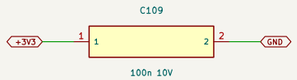


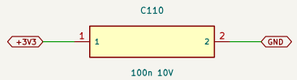












Q=−10k*22n/40*(V_end−V_start); 600nC=109.091mV nominal. Take BASELINE, CATH, ANODE, NET/TAIL endpoints.

Cathodic and anodic charges remain separate. Opposed phases can cancel: final net alone cannot qualify delivery.

Actual current−tail/window evidence is required. Command edges, the comparator or an ideal16−bit LSB do not certify precision.

CLRN = HI_OK AND LO_OK AND EXT_SUP_OK. Any low latches Q off; rising REARM admits only while CLRN is high.

ARTIFICIAL LOADS ONLY; native schematic slice, not a completed implant

Project ATO

Sheet: /Source0_monitor/

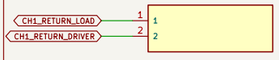
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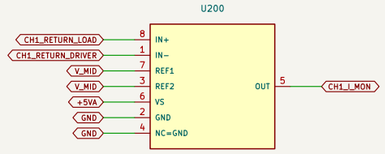
Title: ATO S4 / Source 0: current and phase charge

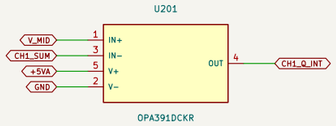
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KiCad E.D.A. 10.0.6		Id: 2/3

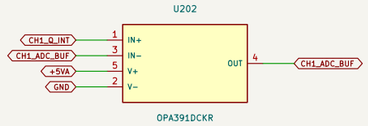
ARTIFICIAL ELECTRICAL LOADS ONLY / FIRST MONITORING SLICE

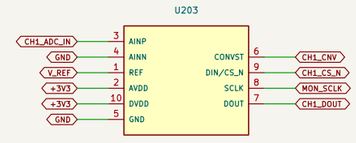
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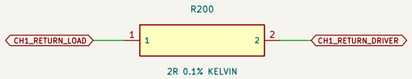




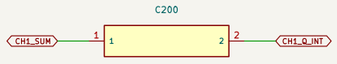




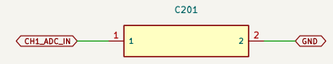


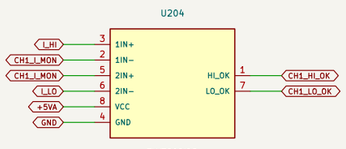


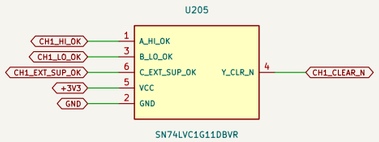


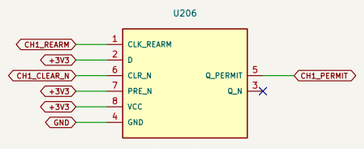


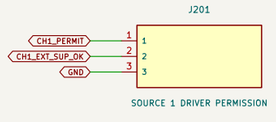


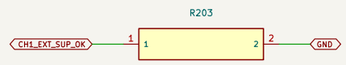
















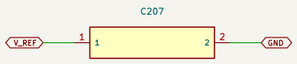


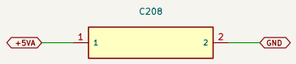


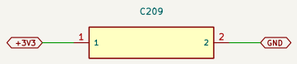


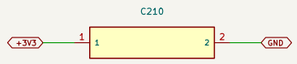












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Project ATO

Sheet: /Source1_monitor/

File: channel1.kicad_sch

Title: ATO S4 / Source 1: current and phase charge

Size: A3	Date: 2026−10−12	**Rev: 0.2 DCK**
KiCad E.D.A. 10.0.6		Id: 3/3